

Problem 2.15

Consider the amount function

$$A(t) = A(0)(1+i)^t.$$

Suppose that a deposit 1 at time $t = 0$ will increase to 2 in a years, 2 at time 0 will increase to 3 in b years, and 3 at time 0 will increase to 15 in c years. If 6 will increase to 10 in n years, find an expression for n in terms of a , b , and c .

Solution

From the first condition,

$$2 = 1(1+i)^a,$$

so

$$2 = (1+i)^a.$$

From the second condition,

$$3 = 2(1+i)^b.$$

From the third condition,

$$15 = 3(1+i)^c,$$

which implies

$$5 = (1+i)^c.$$

Now multiply the first two relations:

$$2(1+i)^a(1+i)^b = 3.$$

Hence,

$$6 = 2(1+i)^a(1+i)^b = 2(1+i)^{a+b}.$$

We are also told that 6 increases to 10 in n years, so

$$10 = 6(1+i)^n.$$

Thus,

$$\frac{10}{6} = (1+i)^n.$$

But from above,

$$\frac{10}{6} = \frac{2(1+i)^c}{2(1+i)^{a+b}} = (1+i)^{c-a-b}.$$

Therefore,

$$(1+i)^n = (1+i)^{c-a-b}.$$

Hence,

$$\boxed{n = c - a - b}.$$